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ICE 6.2

Aman Gupta • Feb 11, 2025 (Edited Feb 21, 2025)

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1. (1 mark) Consider GF(2^4), and arithmetic over its elements using the irreducible polynomial GF(2^4): m(x) = x^4 + x + 1.Which of these elements are NOT a member of GF(2^4): 0010, 00011, 010? Pick all those not in GF(2^4).

- 0010, 00011, 010
- **00011, 010 (correct)**
- 010
- None of the above

2. (1 mark) Refer to question 1. Is it true that 0010 * a = a + a. Here a is any element in GF(2^4).

- Yes
- **No (correct)**
- Maybe

3. (2 marks) Refer to question 1 again, above. What is 0011 + 0011?

- 0022
- 1100
- **0000 (correct)**
- None of the above

4. (4 marks) Refer to question 1 again. What is 0011 * 1100? Recall m(x) = x^4 + x + 1.

- 1111
- 100100
- **0111 (correct)**
- None of the above

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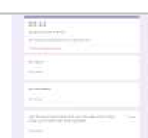

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